Local systems and D-modules

Def^n (Betti local systems) X topological space, G group

G-bundle on X is a sheaf of G-torsors (principal G-bundles)
G top. topological group.

G-local system is a G-bundle $\mathcal{L}_r$

parallel transport: for every path $\gamma : x \to y$, $x, y \in X$

have a map $\mathcal{L}_\gamma : \mathcal{L}_x \longrightarrow \mathcal{L}_y$ such that

(a) if $\gamma \underset{h.e}{\simeq} \gamma'$, then $\mathcal{L}_\gamma = \mathcal{L}_{\gamma'}$

(b) if $x \xrightarrow{\gamma} y \xrightarrow{\gamma'} z$ then

$$\mathcal{L}_{\gamma' \circ \gamma} = \mathcal{L}_{\gamma'} \circ \mathcal{L}_\gamma \qquad \left(\overset{?}{\mathcal{L}_{id}} = id \right)$$

Rmk: X scheme w/ Zariski topology NOT a good example.

Better example: $X^{an} = X(\mathbb{C})$

Rmk: $G = GL_n(k)$, then this is equivalent to k-bundles via Borel construction. $V = k^n$ "standard rep"

$$\begin{array}{c} P \\ \downarrow \\ X \end{array} \text{ G-torsor} \rightsquigarrow \begin{array}{c} P \times^G V \overset{def}{=} \dfrac{P \times V}{G} \\ \downarrow \text{ v.b.} \\ X \end{array} \qquad g \cdot (p, v) = (pg, g^{-1} v)$$

Exercise: $G = SL_2$ Come up w/ vector bundle description
G: finite closed group

Remark: The parallel transport maps are $\simeq$.

Exercise: $\left\{ \begin{matrix} G\text{-} \\ \text{local sys} \\ \text{on} \\ X \end{matrix} \right\} \longleftrightarrow \left\{ \begin{matrix} \text{group} \\ \text{hom} \end{matrix} \quad \pi_1(X) \to G \right\}$.

Question: Can we formulate this in algebraic geometry?

i.e. X variety $\rightsquigarrow$ $X(\mathbb{C})$ local systems on

Yes: Need "de Rham" local system.

Def: k field, alg. cl., char 0 $= \mathbb{C}, \overline{\mathbb{Q}}_\ell$

X smooth variety $/k$

E vector bundle on X w/ sheaf of sections $\mathcal{E}$

A <u>connection</u> on $\mathcal{E}$ is $\underline{\underline{k}}$-linear map

$\nabla : \mathcal{E} \to \Omega^1_X \otimes_{\underline{\underline{k}}} \mathcal{E}$ satisfying Leibniz rule:

$$\boxed{\nabla(f\sigma) = f\nabla(\sigma) + df \otimes \sigma}$$ $\leftarrow$ not $\mathcal{O}_X$ lin

Rmk: comes from differentiable manifolds, complex manifolds

Idea: talk how relating "infinitesimally close" fibers

Replace:
- evaluate at fixed vector fields Θ
- take some duals ($\mathcal{O}_X$ lin duals ...)

Def If Θ is a <u>vector field</u> then the <u>covariant derivative</u>

$\nabla_\Theta \in \mathrm{End}_k(\mathcal{E})$

Leibniz rule: $[\nabla_\theta, f] = \theta(f)$

Because... $\mathcal{O}_X$ vs k dual... extra condition

$$\nabla_{f\theta} = f\nabla_\theta \quad \text{OR} \quad \nabla_{f\theta} = \nabla_\theta f$$

two versions... why?

$\mathcal{O}_X$ as mult.

$\mathcal{E}nd_k(\mathcal{E})$ is $\mathcal{O}_X$-$\mathcal{O}_X$ bimodule.

post precomposition

[Contrasts: $\mathcal{E}nd_{\mathcal{O}_X}(\mathcal{E})$ $\mathcal{O}_X$-module.]

Why: $\nabla : \mathcal{E} \longrightarrow \Omega^1_X \otimes_k \mathcal{E}$.

$$\begin{tikzcd} T_X \otimes_{\mathcal{O}_X} \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{E}^\vee \arrow[r] & T_X \otimes_{\mathcal{O}_X} \Omega^1_X \otimes_k \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{E}^\vee \\ T_X \arrow[u] & \mathcal{E} \otimes \mathcal{E}^\vee \arrow[u] \end{tikzcd}$$

$\{$ ev. at θ

get k-lin map...

$$\nabla' : T_X \longrightarrow \mathcal{E}nd_k(\mathcal{E})$$

Want: intuitive // transport via connections...

Def: The curvature $K \in \Gamma(X, \Omega^2_X \otimes_k \mathcal{E}nd_k(\mathcal{E}))$

is $K(\theta_1, \theta_2) = \nabla_{\theta_1}\nabla_{\theta_2} - \nabla_{\theta_2}\nabla_{\theta_1} - \nabla_{[\theta_1, \theta_2]}$

idea: $[\theta_1, \theta_2]$ measuring the diff. between

flowing in θ_1, then θ_2

vs. θ_2 then θ_1

Def^n The connection ∇ is <u>flat</u> if $\kappa = 0$.

Question: How can we formulate this more algebraically? i.e. w/o reference to vector fields?

Def A <u>derivation</u> on smooth scheme X is an element $D \in \mathrm{End}_k(\mathcal{O}_X)$ satisfying Leibniz rule:

$$D(fg) = f D(g) + D(f) g.$$

These assemble into a k-linear sheaf Der_X $\mathcal{O}_X$-$\mathcal{O}_X$-module sheaf. via composition w/ mult.

Exercise: Define T_X on X two equivalent ways:

① $T_X := \mathrm{Der}_X$

② $\mathcal{O}_X$-linear dual of Ω^1_X ← define this $\mathcal{I}_\Delta / \mathcal{I}_\Delta^2$, $\mathcal{I}_\Delta = X \times X$ — sequence of chain of subdiagonals.

what happened to two-subdiagonals?

Def^n X smooth scheme.

The <u>sheaf of diff. op</u> on X $\mathcal{D}_X \subseteq \mathrm{End}_k(\mathcal{O}_X)$ ← not quasicoh... ~~not quasicoh~~

is the subalgebra generated by $\mathcal{O}_X$ ← multiplication

and T_X ← derivations

- sheaf of noncommutative algebras
- $\mathcal{O}_X$-$\mathcal{O}_X$ bimodule.

} inherited from $\mathrm{End}_k(\mathcal{O}_X)$

Define an <u>order filtration</u>: $\mathcal{D}_X^{\leq 0} := \mathcal{O}_X$, $\mathcal{D}_X^{\leq k} = \{ \sigma \in \mathcal{D}_X \mid [\sigma, f] \in \mathcal{D}_X^{\leq k-1} \text{ for all } f \in \mathcal{O}_X \}$

Remark: any "word" $f_1\theta_1 f_2\theta_2 \cdots \theta_r f_r$

functions, derivations

determines an element in D_X, every element has a representation $\sum$ words (not unique)

Claim: $[\theta, f] = \theta(f)$ Leibniz rule.

Claim: (Exercise) Hint: Jacobi identity. | Order filtration is ...

(a) "strongly" multiplicative $D^{\le i} D^{\le j} = D^{\le i+j}$

(b) $[-,-]$ decreases order by one, i.e.

$$[-,-] : D^{\le i} \otimes D^{\le j} \longrightarrow D^{\le i+j-1}$$

(c) exhaustive: $D_X = \bigcup_{i=0}^{\infty} D_X^{\le i}$

(d) each $D^{\le i}$ is locally free of finite rank.

(e) $gr_k(D_X) \simeq \mathrm{Sym}^k_{\mathcal{O}_X} T_X$ $\rightsquigarrow$ $\mathcal{O}_X$-basis of D_X

i.e. T_X locally free, so locally, choose basis $\theta_1, \dots, \theta_n$

locally, basis for D_X is monomials in $\theta_1, \dots, \theta_n$.

think about $\mathrm{Mod}(D_X)$ as "noncommutative deformation" "quantization" of $\mathrm{QCoh}(T^*_X)$

Prop: $D_X \simeq \mathrm{Sym}^\bullet_{\mathcal{O}_X} T_X$, i.e. D_X quasicoherent as $\mathcal{O}_X$-module on either side.

Pf: $0 \to \mathcal{O}_X \to D^1_X \to T_X \to 0$ splits

Ex: $X = \mathbb{A}^1 = \operatorname{Spec} k[x]$ $\quad \Gamma(X, D_X) = \dfrac{k\langle x, \partial_x \rangle}{[\partial, x] = 1}$ $\quad [\partial_x = \frac{d}{dx}]$ Weyl algebra

$\frac{d}{dx}(xf) - x \cdot \frac{d}{dx} f = f$

D_X is $\mathcal{O}_X$-quasicoherent, "not an algebra"

$$D_X(\mathbb{A}^1) = \frac{k\langle x, x^{-1}, \partial \rangle}{[\partial x] = 1} \cong \frac{k\langle x, \partial \rangle}{[\partial x] = 1} \otimes_{k[x]} k[x, x^{-1}]$$

$\mathbb{A}^1 = \mathbb{G}_m$ $\qquad [\partial x^{-1}] = -x^{-2}$

Exercise: If $X = k[x,y]/xy$, show D_X is not finitely generated.

Def: left D_X-module is a $\mathcal{O}_X$-quasicoherent sheaf $\mathcal{M}$ of D_X-sheaf of D_X left-modules that is $\mathcal{O}_X$-quasicoherent.